

Robert C. Viesca (August 2025)

CONTACT	207 Anderson Hall 200 College Avenue Medford, Massachusetts, 02155, USA	<i>website:</i> viesca.github.io <i>e-mail:</i> robert.viesca@tufts.edu
ACADEMIC POSITIONS	Tufts University , <i>Department of Civil and Environmental Engineering</i> , Medford, MA Research Assistant Professor, 09/11–08/12; Assistant Professor, 09/12–08/18; Associate Professor, 09/18–present Dalhousie University , <i>Department of Civil and Resource Engineering</i> , Halifax, NS, Canada Postdoctoral Fellow with Prof. Dmitry I. Garagash, 09/11–08/12	
VISITING POSITIONS	University of Tokyo , <i>Earthquake Research Institute</i> , Tokyo, Japan Visiting Professor, 09/25–06/26, host: So Ozawa GeoAzur , <i>Observatoire de la Côte d’Azur</i> , Sophia Antipolis, France Chercheur Invité, 06/19–07/19, hosts: Frederic Cappa, Jean-Paul Ampuero MINES ParisTech , <i>Centre de Géosciences (Géophysique)</i> , Fontainebleau, France Professeur Invité, 06/16–07/16, host: Pierre Dublanchet Institut de Physique du Globe , <i>Tectonique et Mécanique de la Lithosphère</i> , Paris, France Professeur Invité, 06/14–07/14; Visitor, 06/13–07/13, 07/15, hosts: Harsha Bhat, Yann Klinger	
EDUCATION	Harvard University , <i>School of Engineering and Applied Sciences</i> , Cambridge, MA Engineering Sciences: S.M. 06/06, Ph.D. 11/11 with Prof. James R. Rice Tufts University , <i>Department of Civil and Environmental Engineering</i> , Medford, MA B.S., Civil Engineering, <i>summa cum laude</i> , 05/05	
PUBLICATIONS	Viesca, R. C. (2025), Asymptotic solutions for self-similarly expanding fault slip induced by fluid injection at constant rate, <i>J. Fluid Mech.</i> , 1019, A27, doi:10.1017/jfm.2025.10593 Ciardo, F., and R. C. Viesca (2025) Non-linear stability analysis of slip in a single-degree-of-freedom elastic system with frictional evolution laws spanning aging to slip, <i>Proc. Roy. Soc. A</i> , 481, 20240917, 2025. doi:10.1098/rspa.2024.0917 Thomas, C., I. Svetlizky, G. Albertini, R. C. Viesca , S. M. Rubinstein, F. Spaepen, C. Yuan, M. Denolle, Y.-Q. Song, L. Xiao, D. A. Weitz (2024) Propagation of extended fractures by local nucleation and rapid transverse expansion of crack-front distortion, <i>Nature Phys.</i> , doi:10.1038/s41567-023-02365-0 Jacquey, A. B. and R. C. Viesca (2023) Nucleation and arrest of fluid-induced aseismic slip, <i>Geophys. Res. Lett.</i> , 50, e2022GL101228, doi:10.1029/2022GL101228 Viesca, R. C. (2023), On the existence of a nucleation length for dynamic shear rupture, <i>J. Mech. Phys. Solids</i> , 173, 105221, doi:10.1016/j.jmps.2023.105221 Sáez, A., B. Lecampion, P. Bhattacharya, and R. C. Viesca (2022) Three-dimensional fluid-driven stable frictional ruptures, <i>J. Mech. Phys. Solids</i> , 160, 104754, doi:10.1016/j.jmps.2021.104754 Viesca, R. C. (2021), Self-similar fault slip in response to fluid injection, <i>J. Fluid Mech.</i> , 928, A29, doi:10.1017/jfm.2021.825 Ray, S., and R. C. Viesca (2019), Homogenization of fault frictional properties, <i>Geophys. J. Int.</i> , 219, 1203–1211, doi:10.1093/gji/ggz327 Viesca, R. C. , and P. Dublanchet (2019), The slow slip of viscous faults, <i>J. Geophys. Res.</i> , 124, 4959–4983, doi:10.1029/2018JB016294 Bhattacharya, P. and R. C. Viesca (2019), Fluid-induced aseismic fault slip outpaces pore-	

fluid migration, *Science*, 364(6439), 464–468, doi:10.1126/science.aaw7354

Viesca, R. C., and D. I. Garagash (2018), Numerical methods for coupled fracture problems, *J. Mech. Phys. Solids*, 113, 13–34. doi:10.1016/j.jmps.2018.01.008

Ray, S., and **R. C. Viesca** (2017), Earthquake nucleation on faults with heterogeneous frictional properties, normal stress, *J. Geophys. Res.*, 122. doi: 10.1002/2017JB014521

Brantut, N., and **R. C. Viesca** (2017), The fracture energy of ruptures driven by flash heating, *Geophys. Res. Lett.*, 44. doi:10.1002/2017GL074110

Viesca, R. C. (2016), Self-similar slip instability on interfaces with rate- and state-dependent friction, *Proc. Roy. Soc. A*, 472(2192), 20160254. doi:10.1098/rspa.2016.0254

Viesca, R. C. (2016), Stable and unstable development of an interfacial sliding instability, *Phys. Rev. E.*, 93(6), 060202(R). doi:10.1103/PhysRevE.93.060202

Platt, J. D., **R. C. Viesca**, and D. I. Garagash (2015), Steadily propagating slip pulses driven by thermal decomposition, *J. Geophys. Res.*, 120, B12200. doi:10.1002/2015JB012200

Viesca, R. C., and D. I. Garagash (2015), Ubiquitous weakening of faults by thermal pressurization, *Nature Geoscience*, 8(11), 875–879. doi:10.1038/ngeo2554

Brantut, N., and **R. C. Viesca** (2015), Earthquake nucleation in intact or healed rocks, *J. Geophys. Res.*, 119, B11518. doi:10.1002/2014JB011518

Viesca, R. C., and J. R. Rice (2012), Nucleation of slip-weakening rupture instability in landslides by localized increase of pore pressure, *J. Geophys. Res.*, 117, B03104. doi:10.1029/2011JB008866

Viesca, R. C. (2011), The near and far of pore pressure during landslide and earthquake ruptures, *Ph.D. thesis* Harvard University, 165 pp.

Viesca, R. C., and J. R. Rice (2011), Elastic reciprocity and symmetry constraints on the stress field due to a surface-parallel distribution of dislocations, *J. Mech. & Phys. Solids*, 59, 753–757. doi:10.1016/j.jmps.2011.01.011

Viesca, R. C., and J. R. Rice (2010), Modeling slope instability as shear rupture propagation in a saturated porous medium, in *Submarine Mass Movements and Their Consequences IV* (proceedings of the 4th Int'l. Symp. on Submarine Mass Movements and Their Consequences, Austin, Texas, 8–11 November 2009), eds. D. C. Mosher et al., R.C. Shipp, L. Moscardelli, J. D. Chaytor, C. D. P. Baxter, H. J. Lee, and R. Urgeles, Springer. doi:10.1007/978-90-481-3071-9_18

Viesca, R. C., E. L. Templeton, and J. R. Rice (2008), Off-fault plasticity and earthquake rupture dynamics, 2. Effects of fluid saturation, *J. Geophys. Res.*, 113, B09307. doi:10.1029/2007JB005530

PROFESSIONAL SERVICE & CONSULTING Manuscript reviews for 25+ journals: *Acta Geotechnica*, *Bulletin of the Seismological Society of America*, *Earth and Planetary Science Letters*, *Earth Planets and Space*, *Geology*, *Geomechanics for Energy and the Environment*, *Geophysical Journal International*, *Geophysical Research Letters*, *International Journal of Engineering Science*, *International Journal of Greenhouse Gas Control*, *International Journal of Numerical and Analytical Methods in Geomechanics*, *Journal of Fluid Mechanics*, *Journal of Geophysical Research*, *Journal of Seismology*, *Nature Communications*, *Nature Geoscience*, *Nature Physics*, *Philosophical Transactions of the Royal Society A*, *Physical Review Letters*, *Proceedings of the National Academy of Science*, *Proceedings of the Royal Society of London A*, *Pure and Applied Geophysics*, *Solid Earth*, *Science*, *Scientific Reports*, and *Tectonophysics*

Ad hoc and panel reviews for publishers, funding agencies, and companies: *American Geophysical Union Books*, *BP America Production Co.*, *Comisión Nacional de Investigación Científica y Tecnológica*, *Elsevier Science and Technology Books*, *Israel Science Foundation*, *National Science Foundation*, and *United States Geological Survey*

Scientific or selection committee for:

MITES summer program, MIT (2014–2017)

6th Intl. Conf. on Coupled THMC Processes in Geosystems (GeoProc), 5-7 Jul. 2017, Paris
Engineering Mechanics Institute Conference 2019, 18-21 Jun. 2019, Pasadena, CA

INVITED
TALKS

11/23: Northwestern University, Dept. of Civil & Environmental Eng., SPREE Seminar
10/23: IRP SlowFaults Workshop
09/23: Tufts University, Dept. of Mechanical Engineering Seminar
09/23: Statewide California Earthquake Center (SCEC), Annual Meeting
06/23: CECAM Workshop, EPFL, 3D Cracks and Crack Stability
06/22: Gordon Research Conference, Flow and Transport in Permeable Media
04/22: Caltech, Mechanical and Civil Engineering Department Seminar
09/21: MIT, Dept. of Earth, Atmospheric, and Planetary Sciences, Geophysics Seminar
06/21: Lab. des Fluids Complexes et Leurs Réservoirs, U. de Pau et des Pays de l'Adour
04/20: MIT, Earth Resources Laboratory, FISH Seminar (postponed)
06/19: GéoAzur, Université Nice Sophia Antipolis and Observatoire de la Côte d'Azur
06/19: Centre International des Sciences Mécaniques (CISM), Advanced School
04/19: Columbia University, Lamont-Doherty Earth Observatory, Geodynamics Seminar
09/18: APEC Cooperation for Earthquake Science (ACES), International Workshop
06/18: GéoAzur, Université Nice Sophia Antipolis and Observatoire de la Côte d'Azur
06/18: Banff International Research Station (BIRS), Workshop on Hydraulic Fracturing
11/16: Weizmann Institute of Science, COST Workshop on Dynamics of Frictional Interfaces
10/16: Harvard University, School of Eng. and Applied Sci., Applied Mechanics Colloquium
09/16: Southern California Earthquake Center, Annual Meeting Workshop
06/16: MINES ParisTech, Centre de Géosciences
04/16: MIT, Earth Resources Laboratory, FISH Seminar
12/15: Georgia Tech, Sigma Xi, Monie Ferst Award Symposium in honor of James R. Rice
11/15: Tufts University, Department of Physics and Astronomy, Condensed Matter Seminar
09/15: Utrecht University, Exp. Rock Def. Laboratory, Modeling Fault Friction Workshop
06/15: ETH Zurich, Institute for Geotechnical Engineering
02/15: Tufts University, Dept. of Mathematics, Computational and Applied Math Seminar
09/14: Princeton University, Department of Geosciences, Solid Earth Brown Bag Series
06/14: École Normale Supérieure, Laboratoire de Géologie
06/14: International Hydraulic Fracturing Summit XI, Schlumberger-Doll, Cambridge, MA
10/13: Brown University, Department of Geological Sciences, Solid Earth Dynamics Seminar
10/13: Caltech, Seismolab, Dix Seminar
10/13: Stanford University, Department of Geophysics, Quake Seminar
07/13: Institut de Physique du Globe de Paris, Séminaires communs Tectonique-Sismologie
05/11: GéoAzur, Université Nice Sophia Antipolis and Observatoire de la Côte d'Azur
04/11: Dalhousie University, Faculty of Engineering
03/11: Tufts University, Department of Civil and Environmental Engineering
03/11: Stanford University, Department of Geophysics
02/11: Northwestern University, Department of Civil and Environmental Engineering
12/10: DUSEL Workshop on Earthquake Rupture Experiments in the Homestake Mine
10/09: Rice University, Department of Earth Science
07/09: U.S. Advisory Committee Meeting for Scientific Ocean Drilling
06/09: Numerical Modeling of Crustal Deformation and Earthquake Faulting Workshop
06/09; 04, 09/10; 4/11: Total S.A. meetings for North American sponsored researchers

HONORS
& AWARDS

2018: Tufts University Center for STEM Diversity Faculty Award
2017–2022: National Science Foundation CAREER Award (Geophysics)
2008–9: Schlanger Ocean Drilling Fellowship
2008: Outstanding Student Paper, Seismology Section, American Geophysical Union
2005: Tufts University Lt. Cmdr. Robert J. Manning Memorial Prize
2005: Tufts University Dept. of Civil and Env. Eng. Earle F. Littleton Scholarship
2004: Tufts University Dept. of Civil and Env. Eng. Cataldo Research Fellowship
2004: Boston Society of Civil Engineers Section/ASCE William P. Morse Award
2003: Society of American Military Engineers Max O. Urbahn Scholarship
2001–10: Bill and Melinda Gates Millennium Scholar (deferred during doctorate)

EXTERNAL SUPPORT	National Science Foundation, Geophysics program, 01/14–01/17, 04/19–03/25 National Science Foundation, CAREER program, 04/17–04/23 Southern California Earthquake Center, 02/13–01/16, 05/17–04/19, 02/22–04/23 United States Geological Survey, Earthquake Hazards program, 01/17–12/17
GRADUATE STUDENTS	Graduate students in geosystems engineering: <i>Parker Aubin</i>, B.S., Mathematics, Boston College; M.S., Tufts University (May 2018) Currently: Geophysicist, Infrasense, Inc., Boston, MA <i>Sohom Ray</i>, B.S., Physics, University of Delhi; M.S., Applied Geophysics, IIT, Roorkee Ph.D. thesis: Earthquake nucleation with heterogeneous physical properties (August 2019) Currently: Assistant Professor, Dept. Earthquake Engineering, IIT Roorkee <i>Lichen Wang</i>, B.Eng, Geotechnical Engineering, China University of Geosciences, Wuhan; M.S., Tufts University (May 2019) Currently: Geotechnical Engineer, GEI Consultants, Boston, MA
POSTDOC. FELLOWS	<i>Pathikrit Bhattacharya</i>, Ph.D., Geophysics, Princeton University Currently: Assistant Professor, School of Earth and Planetary Sciences, NISER <i>Federico Ciardo</i>, PhD. Engineering Mechanics, EPFL Currently: Assistant Professor, Northwestern University <i>Antoine Jacquey</i>, Ph.D., GFZ, RWTH Aachen University Currently: Assistant Professor, Polytechnique Montréal
TEACHING	<i>Undergraduate:</i> ES 5: Statics and Dynamics (falls: 2012–2014, 2016–2017) ES 7: Thermodynamics (springs: 2023–present) ES 8: Fluid Mechanics (falls: 2019–present) CEE 12: Introduction to Hydraulic Engineering (springs: 2015–2018, 2020–2023, 2025) <i>Graduate:</i> CEE 142: Advanced Soil Mechanics (springs: 2014, 2016) CEE 194E: Mechanics of the Natural Environment (spring: 2013) CEE 245: Geomechanics (springs: 2017, 2020)
UNIVERSITY & DEPARTMENT SERVICE & COMMITTEES	<i>School of Engineering committees:</i> academic standing: 09/12–08/14, 09/15–present; chair 09/18–present graduate studies and research: 09/14–05/15, 09/19–08/24 <i>Department of Civil and Environmental Engineering committees:</i> graduate program: 09/14–08/24; interim chair 09/14–01/15; chair 09/19–08/24 undergraduate program: 09/12–08/14; BSCE program director, 09/24–present undergraduate advising: 03/13–05/17, 09/19–present